

Smart Farm Management System

Mission

To enhance farm efficiency and animal welfare by automating monitoring and management through IoT and smart technologies.

Vision

To create fully connected, data-driven farms that minimize losses, improve productivity, and support sustainable agriculture.

Values

Innovation Using technology to solve real agricultural challenges. Reliability
, Ensuring consistent, real-time monitoring and response.

Problem Statement: Impact of Human Delay & Risks

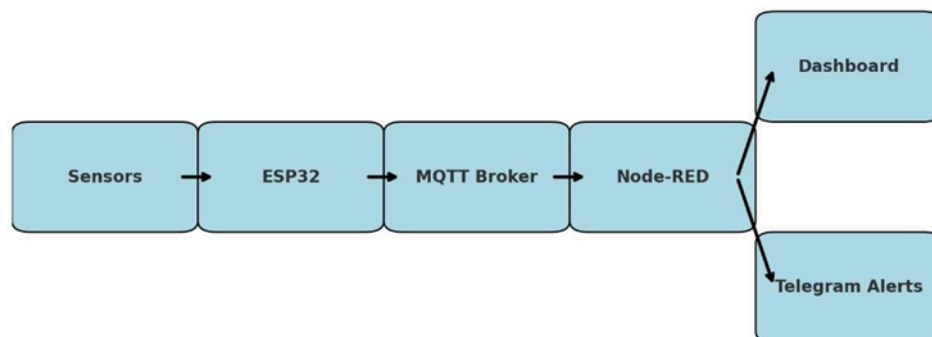
- **One-hour delay in feeding or watering** animals may seem small, but in poultry / intensive livestock operations this can have serious effects:
 1. **Reduced feed conversion efficiency:** Animals deprived of feed or water lose appetite, growth slows, and feed conversion ratios worsen → more feed cost per kg of growth.
 2. **health risks:** Even short periods without water or feed weaken immune systems, increase disease susceptibility. It can reduce weight gain and increase mortality.
 3. **Behavioral issues:** Animals may compete more aggressively, leading to injuries, pecking, or uneven growth.
 4. **Manual task assignment:** Attendance tracking and task assignment are still done manually, wasting time and causing **inefficiency and miscommunication**.
 5. **Health monitoring:** The weight of chickens, a key health indicator, is rarely tracked continuously, which delays identifying health or nutrition problems.

6. **Temperature and gas hazards:** When harmful gases rise or the temperature becomes too high, farmers might not notice immediately — causing **injuries or death** among animals.

- **Human dependency & error:**

- Responsible persons may forget or delay crucial tasks, especially in large farms or during off hours.
- Monitoring manually is costly, error-prone, and difficult to scale.
- Lack of record-keeping leads to no accountability when losses occur.

Smart Farm System Data Flow



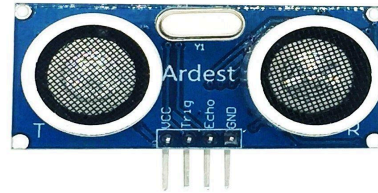
Our Solution :

Our IoT-based Smart Farm System automates monitoring and management processes to create a safer, more efficient, and data-driven farm environment.

Gas and temperature sensors continuously monitor environmental conditions. When unsafe levels are detected, instant alerts are sent to Telegram , allowing quick action before animals are harmed.



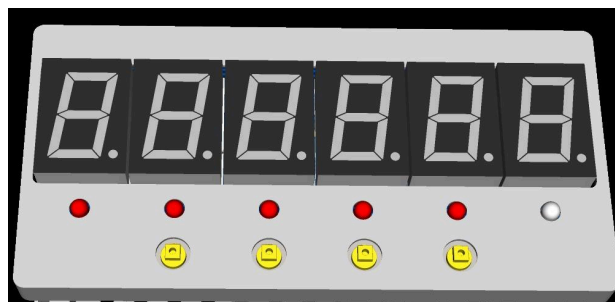
Ultrasonic sensors measure the feed level in real time. When food is low, the system notifies workers automatically, preventing starvation.



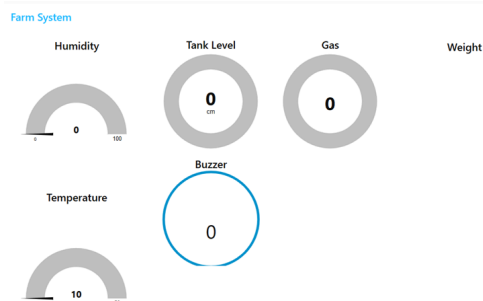
Automated attendance and task assignment: Each worker scans their ID, and the system sends their task directly via email with a “Confirm Done” button — reducing human supervision time.



Weight measurement : A load cell-based system measures chicken weight automatically, helping monitor growth and animal health.



Central dashboard: All readings are displayed on a real-time dashboard, giving the farmer a clear picture of the farm’s condition anytime, anywhere.



The system transforms daily farm operations from manual to fully connected and responsive, saving time, reducing risks, and increasing productivity.

Market Insights & Impact

According to FAO reports / local agriculture studies, **over 20% of livestock losses** are caused by **poor monitoring and delayed response** to environmental conditions.

Manual feeding and attendance tracking consume around **30–40% of workers' time** on small farms.

IoT-based automation can reduce **labor costs by 25%** and **losses by 30%**, while increasing production efficiency.

Our prototype shows that **Telegram alerts reduced response time by more than 70%**, and automated task assignment **eliminated the need for manual coordination**.

System Components

1 Sensors & Devices

- **Temperature & Humidity Sensors (DHT11):** Monitor environmental conditions inside the farm.
- **Ultrasonic Sensor:** Tracks food and water levels to ensure animals are supplied adequately.
- **Gas sensor :** detect ammonia/CO₂/VOCs that indicate poor air quality or risk to workers or animals.
- **(Weight Sensor + HX711):** Measures and records feed or production weights.
- **RFID Module:** Used by workers to log attendance and trigger task assignments.
- **TFT Display:** Provides local, on-site visualization of sensor readings and system status.

Data Flow & Processing

Data Collection (Sensors Layer):

The system starts with multiple sensors placed around the farm

Each sensor continuously collects real-time data from the farm environment.

Data Transmission (Communication Layer):

The sensors are connected to a microcontroller ESP32.

The controller sends the collected data to the Node-RED server using the MQTT protocol, which ensures lightweight and fast communication.

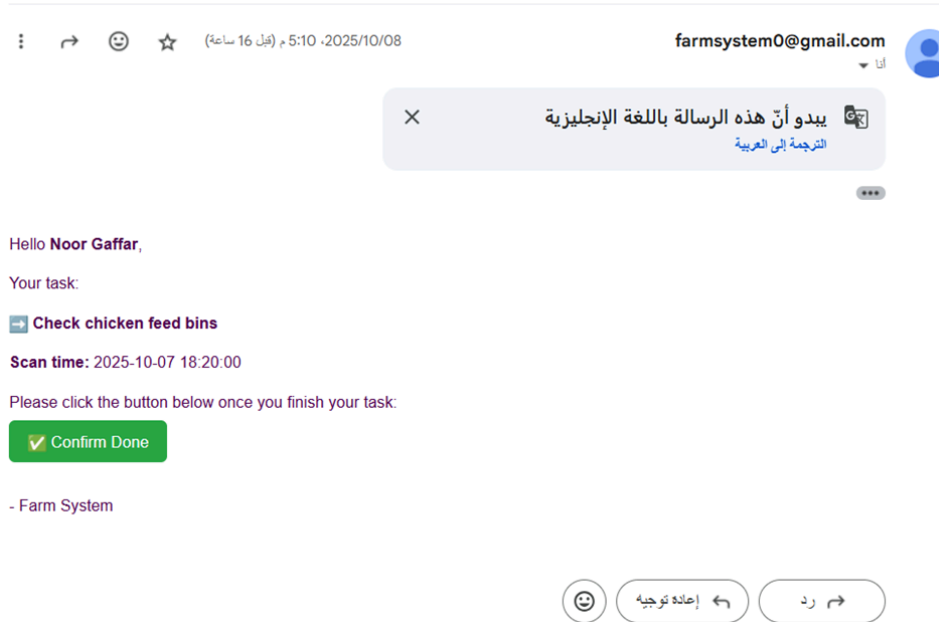
Data Processing (Logic Layer – Node-RED):

Once the data reaches Node-RED, it is processed through logic functions.

The system checks if any value crosses the preset thresholds:

- If gas levels are high → trigger Telegram alert.
- If temperature or humidity is outside the safe range → send Telegram warning.
- If the food level is low → send notification to refill.

The system also records sensor readings and worker attendance data for further analysis.



Task Assignment & Attendance :

- When a worker scans their ID, the system verifies it against the stored data (`workers.json`).
- The system then sends an email containing the worker's assigned task.
- Once the worker finishes the task, they click a "Task Completed" button in the email.

- Node-RED logs this information and stores it in a CSV file for attendance tracking.



column3	column2	column1
07/10/2025 18:20	1234	Noor Gaffar
07/10/2025 18:20	1234	Noor Gaffar
07/10/2025 18:20	1234	Noor Gaffar

Data Storage :

The system stores key data in different formats:

- CSV files → Attendance of workers.
- JSON files → Worker information.

- Dashboard Database → Real-time sensor values.

Visualization (Dashboard Layer):

- A Node-RED Dashboard displays live graphs and indicators for all sensors.
- Farmers can see temperature, humidity, gas levels, food levels, and chicken weights in one place.
- **TFT Screen:** On-site display for quick access to data without requiring a PC or mobile device.

Alert & Response (Action Layer):

1. Whenever an abnormal condition is detected (high gas, low food, temperature issue), the system sends instant Telegram alerts.
2. The farmer can then take action immediately — minimizing risks and improving response time.
3. All sensor data is transmitted through the **MQTT broker (HiveMQ)**.
4. **Node-RED** subscribes to the data topics, processes information, and provides:
 - Real-time visualization dashboards.
 - Automated logic for alerts and task management.
 - Data storage into local files for attendance tracking.



Purpose of Alerts: The Telegram alert is designed to ensure instant communication between the system and farm managers. even short delays can cause significant losses. manual checks is risky, but Telegram ensures real-time, mobile notifications that reach the responsible person wherever they are.

Key Features & Benefits

- Instant Delivery: Telegram messages arrive faster than SMS or email.

- Group Messaging: Alerts can go to multiple staff members simultaneously (workers + supervisors).
- Two-Way Interaction: Workers can respond via Telegram (e.g., confirm they solved the problem).
- Cost-Effective: Free to use, unlike SMS gateways.
- Customizable Severity Levels: Minor warnings vs. critical alerts can have different formats (colors, emojis, urgency).
- Escalation: If the first person does not respond, the system can notify a manager after a set time.

Target Users

Our Smart Farm Management System is designed for **farmers, agriculture managers,** and **livestock producers** who aim to modernize their operations through IoT automation and real-time monitoring. It can also serve as a scalable model for a **factory system**

Future Enhancements

1. AI-Based Predictive Analytics

- Implement a machine learning model to analyze temperature, humidity, gas, and weight data.
- Predict anomalies such as possible diseases, feed shortages, or growth issues before they occur.

2. Automated Control Systems

- Integrate actuators such as automatic feeders, fans, or sprayers that respond directly to sensor data without human input.

3. Cloud and Mobile Integration

- Sync data to a cloud dashboard for remote access via mobile app.
- Provide reports and analytics accessible anywhere for managers and farm owners.